Electric Vehicle Market Performance & Insights

Testing and Quality Assurance

Repository: https://github.com/manasgite54-cmd/EV-project

Purpose and Context

The following cases define the quality assurance plan. Each must be executed and recorded before it can be marked Pass; this document does not claim that unexecuted tests succeeded.

Detailed Project-Specific Content

- T-01 CSV load: read the processed CSV; expected output count is 299,999, subject to verification.
- T-02 Report reconciliation: parse the JSON and compare its output count with the CSV.
- T-03 Schema: compare actual columns/types against the data dictionary.
- T-04 Missingness: calculate null rates for fields used by each chart.
- T-05 SQLite: open/create the database, inspect tables and run representative SELECT/GROUP BY queries.
- T-06 Cross-check: compare a grouped result from SQLite and Pandas using identical data and filters.
- T-07 Flask routes: request Overview, Dashboard, Story and Methodology; check intended response and no server error.
- T-08 Tableau URLs: open Dashboard and Story directly and through the site.
- T-09 Embed fallback: confirm a direct link is visible if the embed is blocked or unavailable.
- T-10 Interaction: test filters, reset state, tooltips and cross-view consistency.
- T-11 Responsive layout: inspect desktop and narrow/mobile widths.
- T-12 Reproducibility: follow setup instructions from a clean environment and record missing dependencies/path assumptions.
- For each case record date, environment, preconditions, steps, expected/actual result, status, evidence and defect reference.
- Allowed statuses: Pass, Fail, Blocked and Not Run. Record security checks for exposed secrets and private data.
- Tester, dates, environment and execution evidence: [TO BE PROVIDED].

Project References and Artifact Locations

Project: Electric Vehicle Market Performance & Insights Repository: https://github.com/manasgite54-cmd/EV-project Processed dataset: data/processed/ev_market_clean.csv Reported processed records: 299,999 Cleaning report: data/processed/cleaning_report.json Tableau Dashboard: https://public.tableau.com/views/Book1_17931030431560/EVMarketOverview?:showVizHome=no&:embed=yes Tableau Story: https://public.tableau.com/views/Book1_17931030431560/Story1?:showVizHome=no&:embed=yes Accuracy note: the record count and artifact paths above are reported by the project owner. Exact schema, source provenance, units, calculated statistics, test results, and deployment status must be checked against the repository and actual artifacts. No unsupported results, approvals, or signatures are claimed.

Status: Detailed project-specific document. Any [TO BE PROVIDED], [TO BE VERIFIED] or [TO BE CALCULATED] item requires confirmation. No supervisor approval, signature, unverified test pass or successful deployment is implied.